COD Guard: Reducing RTO Without Rejecting Good Customers

Founder memo | Based on the supplied April–September 2026 COD order sample

Recommendation

Do not solve RTO by blanket-cancelling risky orders or blocking whole pincodes. Use a risk score to choose a proportionate next step: dispatch low-risk orders, confirm uncertain orders, offer a small prepaid incentive where risk is elevated, and reserve holds for cases with very high risk plus serious contact/address problems. Protect first-time buyers from automatic cancellation. This gives Operations a way to reduce avoidable returns while giving Marketing a chance to save a new-customer relationship.

What the sample actually says

The CSV contains 2,018 rows. After normalizing status text, removing 30 repeated order IDs, and excluding 1 non-final IN_TRANSIT row, the analysis has 1,987 unique orders: 1,422 delivered and 565 RTO. The observed RTO rate is 28.4%.

1. First orders are not automatically safe. Orders with no previous order history had a 30.2% RTO rate (n=1,495), compared with 23.0% for repeat customers (n=492). But first-time buyers still matter for growth: this is a reason to confirm rather than cancel them.

2. A history of repeated RTO is a meaningful warning. Customers with two prior RTOs had a 76.5% observed RTO rate (n=34). The group is small, so it should trigger extra verification, not a permanent customer or pincode ban.

3. Acquisition channel and courier rates vary. Meta Ads had a 31.2% RTO rate (n=765); direct orders were lower at 22.9%. Xpressbees was 30.9% (n=391) and Delhivery was 26.0% (n=699). These are associations, not proof that a channel or courier causes RTO.

4. Geography needs a volume check. Among cities with at least 30 orders, Gorakhpur had the highest observed rate at 62.2% (n=45). This minimum-volume screen avoids overreacting to tiny groups. It is an investigation lead, not enough evidence to block a city or pincode.

Recommended policy and money

The prototype uses a Random Forest to estimate RTO probability, then applies editable thresholds: below 18% ship as-is; 18–32% request WhatsApp confirmation; 32–48% offer prepaid; above that, ask for confirmation or hold for review. Automatic hold is reserved for a stricter 78%+ risk combined with serious data-quality flags. First-time buyers are protected by default. Thresholds are adjustable in the dashboard sidebar without code changes.

On a ■649 reference order, a 2% prepaid offer costs ■12.98. The assignment's assumed COD RTO rate (28%) versus prepaid rate (4%) implies a 24 percentage-point reduction. At ■169.40 direct cost per RTO (■65 forward freight + ■65 return freight + ■25 handling + ■14.40 expected damaged stock), expected RTO cost avoided is ■40.66 per converted order. That leaves an indicative **■27.68 net benefit** before payment fees, discount caps, conversion effects, and margin changes. I recommend 2% capped at ■30.

The assignment's operating scale implies about 1,650 COD orders/month (55% of 3,000), approximately 462 COD RTOs at 28%, and about **■78,263** in direct monthly RTO cost under the stated assumptions. This is a scenario estimate, not a measured figure from the six-month sample.

How the policy would have treated the latest sample month

September 2026 has 329 final-outcome orders in the cleaned sample. A chronological replay using April–August to fit the model and September as the later period assigns 219 to prepaid nudge, 109 to confirmation, and 1 to hold/cancel under the default rules. This is a retrospective policy simulation, not evidence that interventions would actually change outcomes. No orders fall into ship-as-is under these conservative default cut-offs; before launch, calibrate thresholds with Operations and test in a controlled pilot.

Model quality and guardrails

A stratified random 75/25 hold-out produced ROC-AUC 0.715, accuracy 73.8%, and RTO recall 49.6% on 497 test rows. This is a modest sample and random split; the score is useful for ranking but is not certainty. The data contains only COD orders, so it cannot independently validate prepaid behavior. Use time-based evaluation and calibration before production.

What could go wrong, and how we would notice

False positives could push genuine customers into extra steps or cancellation; monitor conversion, first-order completion, complaint rate, and cancellation reversals by action. False negatives could still ship likely RTOs; track RTO rate by risk band, city, courier, and acquisition source. Address language or formatting may be misread; sample flagged addresses for human review.

Courier mix and campaign behavior can change; monitor score distribution and monthly model performance. Never block a pincode on a small sample; require minimum volume and confidence checks.

Next two weeks

Connect Shopify order webhooks and persist every recommendation and override; integrate WhatsApp Business templates with consent and delivery receipts; add a human review queue and payment-gateway verification; run an A/B pilot for confirmation and the 2% capped offer; then compare contribution margin, RTO cost, conversion, and customer complaints. Keep an audit trail so every action can be explained.

Prepared from the supplied assignment brief and CSV. All savings and intervention outcomes are estimates until tested against live operational data.